
Event Generation with Parallel Langevin Sampling and Learned Stein Diagnostics

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Abstract

Efficient event generation is a major computational challenge for precision collider simulations, especially for high-multiplicity final states where matrix-element evaluations are expensive and rejection-sampling efficiencies are low. We study an alternative approach based on many parallel underdamped Langevin chains, retaining one terminal state from each chain to obtain unweighted events while avoiding within-chain autocorrelation. A learned Stein discrepancy is used as a convergence diagnostic, providing a data-driven estimate of the relaxation time. We apply the method to tree-level $u\bar{u} \rightarrow Z + n g$ event generation and find that relaxation requires only a modest number of exact-target Langevin steps, with mild growth over the multiplicities studied. Finally, we show that simple neural-network surrogate initialization can substantially reduce the required number of exact matrix-element and gradient evaluations.

1 Introduction

Monte Carlo event simulation is central to particle physics, connecting theoretical predictions with experimental measurements. The high-luminosity upgrade to the Large Hadron Collider (LHC) [1] and its potential successors [2, 3] will demand unprecedented precision and correspondingly large simulated samples, whose production is expected to become a dominant computational bottleneck [4–8].

The most expensive component of the simulator chain [9] is the sampling of the hard proton-proton collision differential cross section

$$d\sigma_{pp \rightarrow X_n} = \frac{1}{\mathcal{F}} \sum_{i,j} f_i(x_1) f_j(x_2) |M_{ij \rightarrow X_n}(\Phi_n)|^2 dx_1 dx_2 d\Phi_n, \quad (1)$$

where the sum runs over the constituents of the proton (quarks, antiquarks and gluons), f are their corresponding parton density functions, $|M_{ij \rightarrow X_n}|^2$ is the squared matrix element, and $\mathcal{F}$ is the incoming partonic flux factor. The integration measure is given by the two partonic momentum fractions $dx_{1,2} \in [0, 1]$ and the n -body phase-space measure $d\Phi_n$, which integrates over the final-state momenta while ensuring momentum conservation. The total sampling dimensionality of this object is $3n - 2$.

For large multiplicities n and for high theoretical precision, the evaluation of the matrix element can be prohibitively expensive. The method used to sample Eq. (1) must therefore be as efficient as possible in terms of evaluations, even if this comes at increased algorithmic cost. Current methods are still largely based on the VEGAS algorithm [10, 11] with many physics-inspired improvements [12–19]. These methods all rely on an adaptive approximation of an unnormalized target density $\tilde{p}(x)$ (the differential cross section) by some density $q(x)$ that is easily sampled. One can then sample from

33 $q(x)$ and assign weights $w(x) = \tilde{p}(x)/q(x)$. To obtain unweighted samples, rejection sampling must
 34 be performed with acceptance probability

$$p_{\text{acc}}(x) = \frac{w(x)}{\max_x w(x)}. \quad (2)$$

35 The unweighting efficiency $\epsilon = \mathbb{E}_{x \sim q} [p_{\text{acc}}(x)]$ then determines the required number of matrix
 36 element evaluations per generated sample. Eq. 2 is very sensitive to the value of $\max_x w(x)$, which
 37 may originate from tiny regions of phase space. Thus, to improve efficiency, one may make use of a
 38 reduced maximum and leave some undesirable weighted events [20]. Nevertheless, for increasing
 39 multiplicities, the unweighting efficiency can still degrade by several orders of magnitude [21].

40 Recent work has used modern machine-learning (ML) architectures such as normalizing flows in
 41 place of VEGAS to improve efficiencies [22–29]. These approaches are however still ultimately
 42 based on rejection sampling and thus inherit its disadvantages. In this work, we propose a different
 43 approach based on parallelized Markov chain Monte Carlo (MCMC) supplemented by an ML-based
 44 convergence diagnostic. MCMC methods have previously been explored for hard-process phase-
 45 space sampling and event generation [30–33]. Our method places particular emphasis on avoiding
 46 within-chain autocorrelations and on reaching MCMC stationarity, i.e. samples being representative
 47 of $\tilde{p}(x)$. Furthermore, we investigate how neural network surrogates of the target distribution can
 48 further reduce the required number of matrix-element evaluations [34–36].¹

49 2 Method

50 We now describe our sampling construction, targeting an unnormalized density $\tilde{p}(x)$ on the unit
 51 hypercube $x \in [0, 1]^d$. To avoid within-chain autocorrelation, we evolve many Markov chains in
 52 parallel on hardware accelerators and retain only the terminal state from each chain. The learned
 53 Stein discrepancy introduced below is used to determine the relaxation time t_{rel} , i.e. the number of
 54 MCMC steps after which the ensemble is representative of the target.

55 2.1 Underdamped Langevin Algorithm

56 Our MCMC framework is designed for rapid convergence under two practical constraints: the sampler
 57 should be straightforward to run over many parallel chains, and it should use expensive target-density
 58 and gradient evaluations sparingly. Sophisticated Hamiltonian Monte Carlo (HMC) methods [37]
 59 like NUTS [38] implement adaptive trajectory lengths which leads to complex control flow and thus
 60 difficult parallelizability. Moreover, HMC typically requires several intermediate gradient evaluations
 61 before a single Metropolis-Hastings (MH) correction is applied. We therefore use an Underdamped
 62 Langevin Algorithm (ULA) [39, 40], which retains momentum between steps while applying an
 63 MH correction after each proposal. Each parallel chain evolves a position–momentum pair using a
 64 leapfrog proposal, a MH correction, partial momentum refreshment, and adaptive step sizes. The
 65 complete update and adaptation scheme is given in Appendix A.

66 2.2 Mirror Langevin Dynamics

67 After the application of a mapping, the phase-space coordinates used for the cross-section integrand
 68 live on the bounded domain $x \in [0, 1]^d$, while ULA is most naturally formulated on $\mathbb{R}^d$. We therefore
 69 perform the MCMC evolution in mirror space [41], in which Langevin dynamics evaluated through a
 70 suitable mirror map retain the convergence properties of the corresponding unconstrained algorithm.
 71 We use the element-wise transform $y_i = \Phi^{-1}(x_i) = \sqrt{2} \operatorname{erf}^{-1}(2x_i - 1)$. The mirrored target density
 72 is the corresponding change of variables

$$\log \tilde{p}_{\text{mir}}(y) = \log \tilde{p}(\Phi(y)) - \frac{1}{2} \|y\|^2 + \text{const.} \quad (3)$$

73 Note that the constant drops out in the score.

¹Code is available in an [anonymized repository](#).

74 2.3 Learned Stein Discrepancy

75 To monitor convergence of the ULA chains, we need a distributional discrepancy that can be evaluated
 76 using samples from the chain and the unnormalized density of the target. The Stein discrepancy
 77 provides such a diagnostic [42, 43]. For a vector-valued critic function $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$, the Stein
 78 discrepancy between a smooth, sampled distribution p and a smooth target distribution q is

$$S(p, q) = \sup_{f \in \mathcal{F}} \mathbb{E}_{x \sim p} [s_q(x)^T f(x) + \nabla \cdot f(x)], \quad (4)$$

79 where $\mathcal{F}$ is a norm-bounded class of differentiable functions for which the expectation over p is finite
 80 and $q(x)f(x)$ has vanishing tails. For sufficiently rich critic classes, this quantity is non-negative and
 81 satisfies $S(p, q) = 0$ if and only if $p = q$ [44].

82 Kernel Stein discrepancies are a popular choice for $\mathcal{F}$ [45–47], but their standard estimators involve
 83 pairwise sums that scale as n^2 in the number of samples n . Since we typically work in a regime with
 84 large numbers of samples, n^2 evaluation is infeasible, and a more natural choice would be one that
 85 exploits this large data size. We thus use the learned Stein discrepancy (LSD) [48, 49], taking $\mathcal{F}$ to
 86 be a class of square-integrable differentiable functions that satisfy the Stein boundary condition, and
 87 representing the critic by a neural network f_ϕ . In practice, square integrability is enforced by adding
 88 a quadratic regularization term to the training objective,

$$\mathcal{L}_\lambda(\phi; p, q) = -\mathbb{E}_{x \sim p} [s_q(x)^T f_\phi(x) + \nabla \cdot f_\phi(x) - \lambda \|f_\phi(x)\|^2]. \quad (5)$$

89 After training, the learned Stein discrepancy is evaluated with the learned critic on a held-out sample,
 90 without the regularization term.

91 2.4 Sharp Boundaries

92 Physical cross sections typically involve sharp phase-space cuts, which restrict the target density to
 93 an allowed region with a hard boundary. The derivation of Eq. 4 involves an integration by parts over
 94 this region, and a boundary contribution will thus remain unless the Stein test function has vanishing
 95 flux through the cut surface [50, 51]. We address this issue by introducing a scalar boundary function
 96 $h(x)$ that is positive inside the allowed region and vanishes on the cut boundary. The Stein operator
 97 is then applied to the vector field hf rather than to f itself [50, 52],

$$S_h(p, q) = \sup_{f \in \mathcal{F}} \mathbb{E}_{x \sim p} [h(x)s_q(x)^T f(x) + h(x)\nabla \cdot f(x) + \nabla h(x)^T f(x)]. \quad (6)$$

98 2.5 Neural Surrogates

99 The sampling strategy described above relies on gradients of the target density. Differentiable matrix
 100 element implementations have recently become available [29, 53–56], enabling the sampling strategy
 101 set out above, but at higher computational expense. This motivates us to consider the use of learned
 102 surrogates to reduce the required number of evaluations.

103 Neural surrogates for scattering amplitudes and matrix elements have been studied extensively [34–
 104 36, 57–71]. In the present setting, one can train a surrogate on a relatively limited set of points without
 105 requiring gradients. Then, chains can be evolved cheaply through the early part of the trajectory using
 106 the surrogate as target, before switching over to a shorter sequence of ULA steps using the exact
 107 target. The surrogate is thus only used to accelerate equilibration, while the final evolution remains
 108 tied to the physical target density. We investigate this strategy in Section 3.

109 3 Results

110 We validate the learned Stein diagnostic on analytically tractable Ornstein–Uhlenbeck targets in
 111 Appendix B.

112 We apply ULA and the LSD diagnostic to sample the cross section for the process $u\bar{u} \rightarrow Z + ng$.
 113 Matrix elements and their gradients are evaluated using MadJax [53] with several modifications for
 114 numerical stability. We use the NNPDF4.0 LO set [72], and implement the PDF and α_s interpolation
 115 in JAX [73] following the LHAPDF interpolation prescription [74]. We set $\sqrt{s} = 13$ TeV and apply
 116 cuts $p_T > 20$ GeV and $|y| < 5$ to all final-state particles and a separation cut $\Delta R > 0.4$ to the

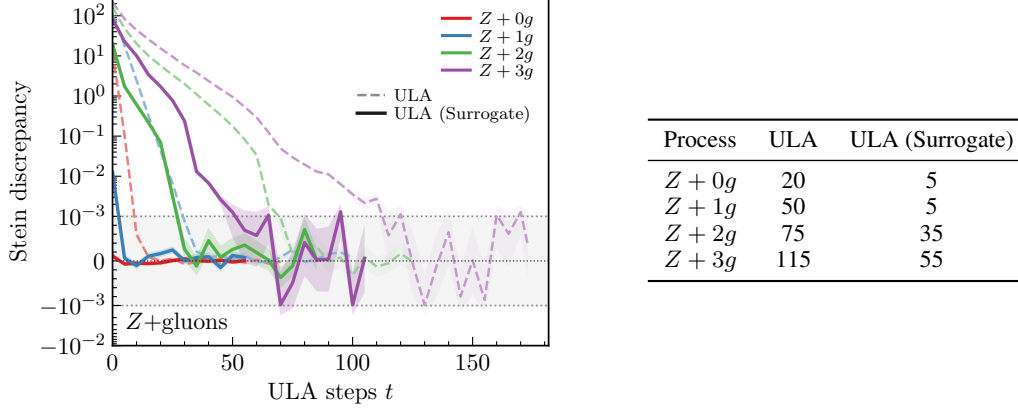


Figure 1: Convergence and relaxation times. Left: the LSD as a function of the number of ULA steps for $Z + ng$ cross sections. Colours denote gluon multiplicities, while line styles distinguish ULA with and without surrogate initialization. Right: relaxation-time checkpoints t_{rel} .

117 gluons. Phase space is mapped to the $[0, 1]^{3n+1}$ hypercube using the Chili [75] vector boson plus jets
 118 map. The Stein diagnostic uses the same critic architecture and optimization strategy as in the OU
 119 validation, with $\lambda = 0.1$. The boundary function is

$$h(x) = \left[\frac{1}{2 + \binom{n}{2}} \left(\frac{1}{1 - x_a} + \frac{1}{1 - x_b} + \sum_{1 \leq i < j \leq n} \frac{1}{1 - \Delta R_{\text{cut}}^2 / \Delta R_{ij}^2} \right) \right]^{-1}, \quad (7)$$

120 for points that pass the cuts, and 0 for points that do not.

121 We use ULA and the LSD diagnostic to generate 10^7 samples for $n = 0, 1, 2, 3$, both with and
 122 without neural surrogates. The surrogate is a residual MLP with five hidden layers of width 128,
 123 GeLU activations, and a scalar output. It is trained to predict the log cross section from only 10^6
 124 samples using Adam and a mean-squared error loss. We then run ULA on the surrogate for a large
 125 number of steps. The terminal states are then used as the initial states for the ULA run, which
 126 is otherwise initialized with unit Gaussian samples. We define the relaxation time t_{rel} as the first
 127 checkpoint for which the learned LSD is no more than one standard error above zero, $S_{\text{LSD}} \leq \Delta S_{\text{LSD}}$.
 128 After the trigger, we evolve the chain for a further $\max(0.5 t_{\text{rel}}, 50)$ ULA steps. Figure 1 shows
 129 the convergence results and summarizes the relaxation-time triggers. The surrogate warm start
 130 substantially reduces the number of exact-target ULA steps required for relaxation. All reported
 131 experiments were run on a single NVIDIA GeForce RTX 5090 GPU (32 GB) and required less than
 132 24 hours in total.

133 Figure 2 compares ULA samples with and without surrogate initialization against independent VEGAS
 134 estimates for four physical observables. The jet-resolution scale $\sqrt{d_{01}}$ is obtained using exclusive
 135 k_t clustering with $R = 1$ [76, 77]. In every case, the terminal ULA samples reproduce the VEGAS
 136 distributions within statistical uncertainties.

137 4 Discussion

138 The results above suggest that gradient-based parallel MCMC is a practically viable alternative to
 139 rejection-based event generation, at least in the regime studied here. The central practical quantity is
 140 the number of exact matrix-element evaluations needed before terminal samples can be used. For
 141 the $Z + ng$ examples, the learned Stein diagnostic triggers after a modest number of exact-target
 142 ULA steps, and the increase with multiplicity is mild over the range considered. This is encouraging
 143 because unweighting efficiencies can degrade by orders of magnitude as the final-state multiplicity
 144 increases [34, 35]. Furthermore, the methods explored here generate strictly unweighted events, while
 145 in the usual rejection sampling setup residual weights may remain when using the maximum weight
 146 would otherwise reduce the efficiency even further. Like conventional rejection-based generation, our
 147 method is also naturally parallel: each chain can be evolved independently, while the adaptation and

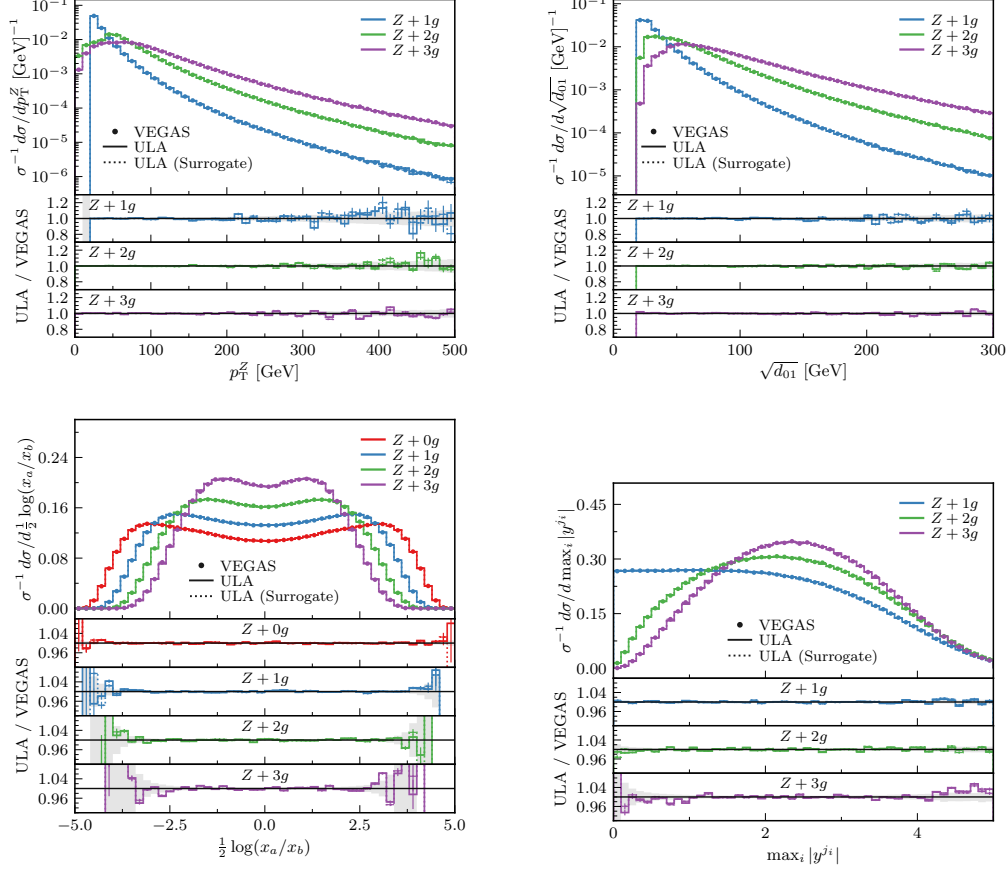


Figure 2: Normalized differential distributions for the Z -boson transverse momentum, the exclusive k_t clustering scale $\sqrt{d_{01}}$, the full-system boost rapidity $\frac{1}{2} \log(x_a/x_b)$, and the largest absolute gluon rapidity $\max_i |y^{j_i}|$. Filled markers show the VEGAS reference, solid lines show ULA samples, and dotted lines show ULA samples after surrogate initialization. The lower panels show the corresponding ratios to VEGAS for each multiplicity.

Stein diagnostic require only ensemble-level reductions. This makes the method well-matched to GPU/TPU execution.

The required number of evaluations can be further reduced through the use of surrogate initialization. In our implementation the surrogate is deliberately simple: it uses the same network architecture as the Stein critic and is trained on a comparatively small set of uniformly sampled phase-space points. Nevertheless, it substantially shortens the exact-target relaxation time. This should be viewed as a conservative demonstration rather than an optimized surrogate strategy. More expressive architectures, improved training samples, or active refinement in the regions explored by the chains could plausibly make the warm start closer to the physical target and further reduce the number of exact matrix-element and gradient evaluations.

Reverse-mode automatic differentiation requires one forward and backward pass, so for tree-level matrix elements composed of rational functions, gradient evaluation should approach a small constant-factor overhead over regular evaluation [78, 79].

The present study considers a fixed partonic channel, tree-level matrix elements, and a simple phase-space map. The observed scaling, and more generally any comparison with current methods, can only be tested in the context of an implementation in full event generator programs. Still, the combination of parallel chains, an explicit relaxation-time diagnostic, and surrogate initialization provides a concrete route towards event generation in which the expensive exact target is evaluated as few times as possible.

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476 A Underdamped Langevin algorithm

477 In this appendix, we describe the Underdamped Langevin algorithm implementation in detail. Each
 478 MCMC chain carries a position x and a velocity v . We assume we have access to the score
 479 $s(x) = \nabla \log \tilde{p}(x)$. For stability purposes the score is regularized component-wise to $\tilde{s}_i(x) =$
 480 $s_i(x)/(1 + \eta_i |s_i(x)|)$, where η_i are the MCMC step sizes [80]. At time step t , a new proposal x', v'
 481 is generated following the leapfrog step

$$\begin{aligned} v_{1/2} &= v_t + \frac{1}{2} \Lambda_\eta \tilde{s}(x_t), \\ x' &= x_t + \Lambda_\eta v_{1/2}, \\ v' &= v_{1/2} + \frac{1}{2} \Lambda_\eta \tilde{s}(x'), \end{aligned} \quad (8)$$

482 where $\Lambda_\eta = \text{diag}(\eta_1, \dots, \eta_d)$. This step discretizes the Hamiltonian dynamics on the augmented
 483 density

$$\pi(x, v) \propto \tilde{p}(x) \exp\left(-\frac{1}{2} v^T v\right). \quad (9)$$

484 In the continuous-time limit, the Hamiltonian flow conserves the Hamiltonian $H(x, v) = -\log \tilde{p}(x) +$
 485 $v^T v/2$ and preserves phase-space volume. As a result, $\pi(x, v)$ as well as its x -marginal, which is
 486 proportional to the desired target density $\tilde{p}(x)$, is invariant. However, at finite step sizes an integration
 487 error is introduced. By applying a MH acceptance step with probability

$$a = \min\{1, \exp[-H(x', v') + H(x_t, v_t)]\} \quad (10)$$

488 the discretization bias is corrected. Upon rejection, the position is kept and the incoming momentum
 489 is reversed $v_t \rightarrow -v_t$. After the MH step, the momentum is refreshed

$$v_{t+1} = \beta v_{\text{acc}} + \sqrt{1 - \beta^2} \zeta, \quad \zeta \sim \mathcal{N}(0, I). \quad (11)$$

490 These momentum updates preserve the Gaussian velocity marginal and inject randomness into the
 491 otherwise deterministic Hamiltonian evolution, allowing the chain to move between trajectories and
 492 explore the full target distribution. After equilibration, terminal positions drawn from these chains
 493 are distributed according to $\tilde{p}(x)$.

494 We write the step sizes as $\log \eta_i = \log \epsilon + \log \rho_i$, with $\sum_i \log \rho_i = 0$. The scalar scale ϵ is
 495 initialized by doubling or halving until the average acceptance probability is close to one half [38],
 496 while the directions ρ_i are initialized from the diagonal score second moment, motivated by Fisher
 497 preconditioning [81, 82],

$$F_{ii,t} = \mathbb{E}_{x_t} [s_i(x_t)^2], \quad \log \rho_i = -\frac{1}{2} \log F_{ii,t} + \frac{1}{2d} \sum_{j=1}^d \log F_{jj,t}. \quad (12)$$

498 This fixes $\prod_i \rho_i = 1$ and increases step sizes in directions with smaller typical score variation.
 499 During the chain, $\log \epsilon$ is adapted by ascending the expected squared jump distance $\text{ESJD}(\epsilon) =$
 500 $\mathbb{E}_{x_t, v_t, \zeta} [a \|x' - x_t\|^2]$ at fixed ρ , a standard proxy for MCMC efficiency [83, 84]. The directions are
 501 adapted by updating ρ_i with new estimates of the diagonal score second moment. Both the ESJD scale
 502 update and the score-moment direction update use the same diminishing Robbins–Monro schedule,
 503 so that the transition-kernel changes vanish asymptotically, as required for stationarity [85, 86]. All
 504 expectations are evaluated over the parallel chain ensemble.

505 B Stein diagnostic validation

506 In this appendix, we validate the Stein diagnostic on a target for which the MCMC convergence can
 507 be computed analytically. Starting from a fixed point μ_0 away from some target Gaussian mean μ_q
 508 and diagonal target Σ_q , we sample an exact Ornstein–Uhlenbeck (OU) transition kernel

$$x_{t+1} = \mu_q + e^{-\eta \Sigma_q^{-1}} (x_t - \mu_q) + \Sigma_q^{1/2} \left(I - e^{-2\eta \Sigma_q^{-1}}\right)^{1/2} \xi, \quad \xi \sim \mathcal{N}(0, I), \quad (13)$$

where η is the step size. After t steps, the marginal distribution is Gaussian with

$$\begin{aligned}\mu_t &= \mu_q + e^{-t\eta\Sigma_q^{-1}}(\mu_0 - \mu_q) \\ \Sigma_t &= \Sigma_q \left(I - e^{-2t\eta\Sigma_q^{-1}} \right).\end{aligned}\tag{14}$$

Thus p_t converges to the target density as $t \rightarrow \infty$, yielding a controlled and testable sequence of distributions. To validate the role of the boundary function, we apply a radial cut that removes 20% of the target probability mass centered around the target mean μ_q .

Under Eq. (5), the optimal Stein discrepancy is [48]

$$S_{\text{opt}}(p_t, q) = \frac{1}{2\lambda} \mathbb{E}_{x \sim p_t} \left[\|s_q(x) - s_{p_t}(x)\|^2 \right].\tag{15}$$

We compare this optimal value against the LSD computed on OU samples and on ULA samples. The validation results use dimensions $d = 4$ and $d = 16$, with target parameters

$$\mu_q = \text{linspace}(-2, 2, d), \quad \Sigma_q = \text{diag}(\text{linspace}(0.5, 2, d)^2).\tag{16}$$

The OU process is initialized from the fixed point $\mu_0 = \mu_q + 3$, while ULA is initialized from a unit Gaussian. The LSD regularization is $\lambda = 0.1$. The Stein critic is modelled with a five-layer residual GeLU network with hidden width 128. Optimization is performed with the Adam optimizer [87] on $n = 10^7$ chains, and final LSD values are obtained using 10-fold cross validation. The boundary function is $h(x) = \max\{1 - r_{\text{cut}}^2/r^2(x), 0\}$.

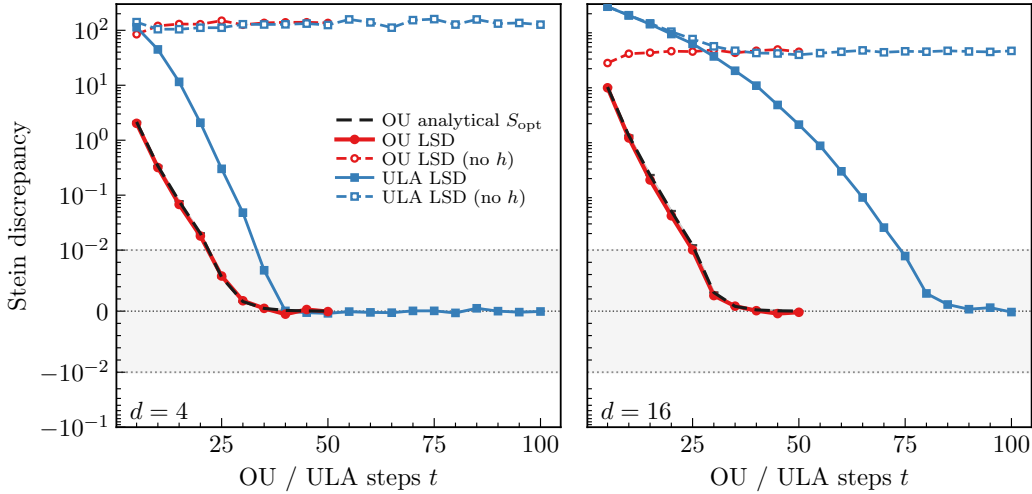


Figure 3: The LSD as a function of the number of steps for the exact OU reference process with $\eta = 0.5$ (red) and the underdamped Langevin algorithm (ULA, blue). Dashed lines show the boundary-function ablation. The black line shows the analytic optimal Stein discrepancy S_{opt} .

The validation results are shown in Fig. 3. For the exact OU samples, the learned LSD tracks the analytic value S_{opt} , supporting its use as a quantitative convergence diagnostic. The ULA curves show similar monotonic decay towards zero, with the higher-dimensional case converging more slowly, qualitatively consistent with the expected $\sqrt{d}$ scaling of Metropolis-adjusted Langevin methods [88]. In the absence of the boundary function, the remaining finite boundary contribution prevents the Stein discrepancy from approaching zero.